

交通大数据

计数回归

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计数回归

□ 学习目标

- 计数数据
- 泊松回归
- 负二项回归
- 零膨胀泊松回归、负二项回归



计数回归

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● 计数数据

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- 负二项回归
- 零膨胀泊松回归、负二项回归



计数数据 (count data)

□ 什么是计数数据?

- 不能连续取值, 只能是0,1,2,3等**非负整数**
 - ✓ 不合格数: 二项分布
 - ✓ 事故数: 泊松分布



计数数据 (count data)

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- 不能连续取值, 只能是0,1,2,3等**非负整数**
 - ✓ 不合格数: 二项分布
 - ✓ 事故数: 泊松分布

□ 二分类或多分类变量 = 计数数据?

- **属于离散数据, 不属于计数数据**
 - ✓ 分类: 名义变量
 - ✓ 计数: 物理意义



计数回归

□ 学习目标

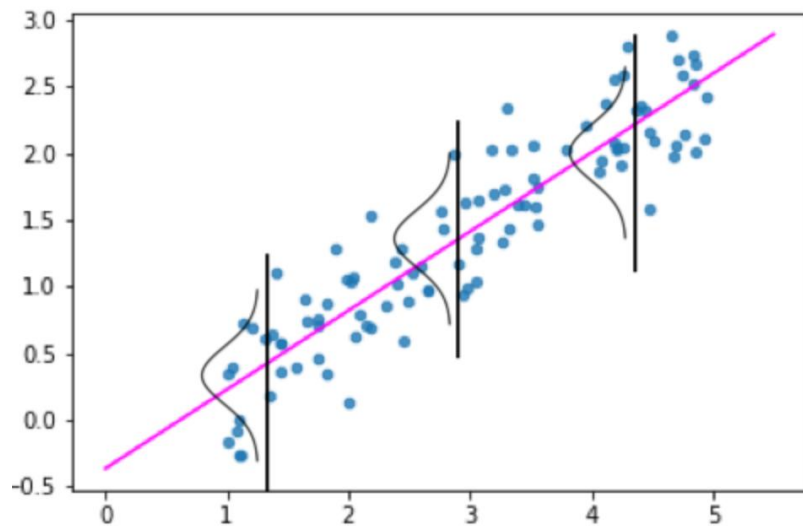
- 计数数据
- **泊松回归**
- 负二项回归
- 零膨胀泊松回归、负二项回归



泊松回归 (Poisson regression)

□ 广义线性模型 (generalized linear model)

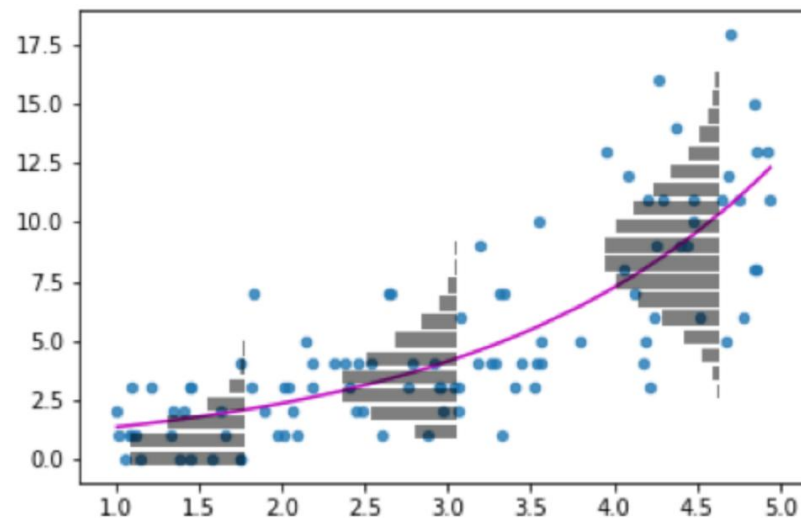
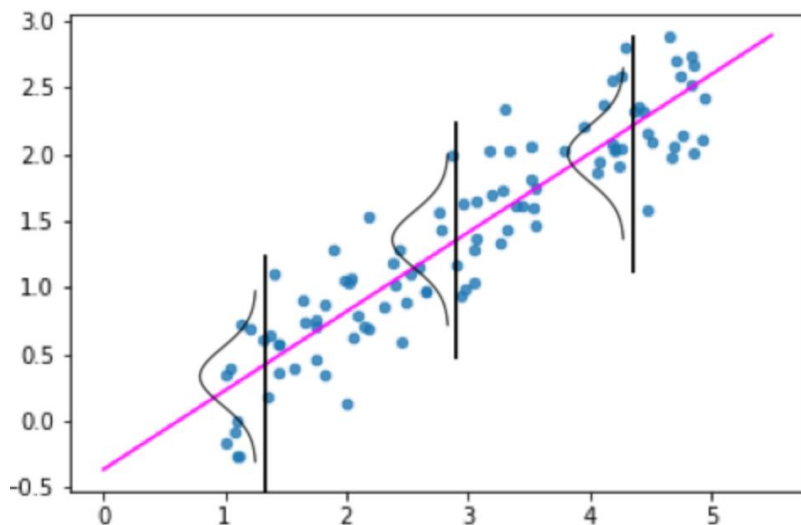
- 线性回归仅考虑因变量服从**正态分布**



泊松回归 (Poisson regression)

□ 广义线性模型 (generalized linear model)

- 线性回归仅考虑因变量服从**正态分布**
- **GLM** 考虑因变量服从**指数族 (exponential family) 分布**
 - ✓ 包括正态分布、泊松分布、二项分布等



泊松回归 (Poisson regression)

□ 广义线性模型 (generalized linear model)

给定观测数据 X 和 Y , GLM 定义如下三个组成部分:

① 随机部分 (random component) : $Y|X$ 的分布

➤ 线性回归假设 $Y|X \sim N(\mu, \sigma^2)$

➤ GLM 假设

$$f(y|\theta, \phi) = \exp\left(\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right)$$

✓ θ : 自然参数 (natural parameter) , ϕ : 离散参数 (dispersion parameter)

✓ a, b, c 是函数

✓ **任何概率密度满足以上形式, 属于指数族分布**

✓ 正态分布: $\theta = \mu, \phi = \sigma$



泊松回归 (Poisson regression)

□ 广义线性模型 (generalized linear model)

给定观测数据 X 和 Y , GLM 定义如下三个组成部分:

② **系统部分 (systematic component)** : 参数 η 与 X 的数学关系

$$\eta = X^T \boldsymbol{\beta} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 \dots + \beta_d X_d$$



泊松回归 (Poisson regression)

□ 广义线性模型 (generalized linear model)

给定观测数据 X 和 Y , GLM 定义如下三个组成部分:

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③ **连接函数 g (link function)** : 把随机部分和系统部分联系起来

$$g(\mu) = \eta$$

➤ $\mu = \mathbb{E}(Y|X)$

➤ 线性回归: $g(\mu) = \mu$



泊松回归 (Poisson regression)

□ 广义线性模型 (generalized linear model)

四个参数 θ , ϕ , μ , η , 梳理一下!

➤ μ 是**估计对象**, 即 $E(Y|X)$



泊松回归 (Poisson regression)

□ 广义线性模型 (generalized linear model)

四个参数 θ , ϕ , μ , η , 梳理一下!

- μ 是**估计对象**, 即 $E(Y|X)$
- θ 是**概率密度函数的自然参数**, 影响分布的位置, 所以 μ 依赖于 θ



泊松回归 (Poisson regression)

□ 广义线性模型 (generalized linear model)

四个参数 θ , ϕ , μ , η , 梳理一下!

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- θ 是**概率密度函数的自然参数**, 影响分布的位置, 所以 μ 依赖于 θ
- ϕ 是**概率密度函数的离散参数**, 估计 μ 不需要知道 ϕ



泊松回归 (Poisson regression)

□ 广义线性模型 (generalized linear model)

四个参数 θ , ϕ , μ , η , 梳理一下!

- μ 是**估计对象**, 即 $E(Y|X)$
- θ 是**概率密度函数的自然参数**, 影响分布的位置, 所以 μ 依赖于 θ
- ϕ 是**概率密度函数的离散参数**, 估计 μ 不需要知道 ϕ
- η 是**观测自变量的线性组合** (linear combination)



泊松回归 (Poisson regression)

$$f(y|\theta, \phi) = \exp\left(\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right)$$

□ 广义线性模型 (generalized linear model)

➤ 正态分布 (normal distribution)

$$f(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y - \mu)^2}{2\sigma^2}\right)$$

$$f(y) = \exp\left(\frac{y\mu - \frac{\mu^2}{2}}{\sigma^2} - \frac{y^2}{2\sigma^2} - \log \sigma - \log \sqrt{2\pi}\right)$$

✓ $\theta = \mu$

✓ $b(\theta) = \frac{\mu^2}{2} = \frac{\theta^2}{2}$

✓ $\phi = \sigma, a(\phi) = \sigma^2, c(y, \phi) = -\frac{y^2}{2\sigma^2} - \log \sigma - \log \sqrt{2\pi}$



泊松回归 (Poisson regression)

$$f(y|\theta, \phi) = \exp\left(\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right)$$

□ 广义线性模型 (generalized linear model)

➤ 伯努利分布 (Bernoulli distribution)

$$f(y) = p^y(1-p)^{1-y}$$

$$f(y) = \exp\left(y \log\left(\frac{p}{1-p}\right) + \log(1-p)\right)$$

$$\checkmark \theta = \log\left(\frac{p}{1-p}\right), \text{ 或 } p = \frac{e^\theta}{1+e^\theta}$$

$$\checkmark b(\theta) = -\log(1-p) = \log(1+e^\theta)$$

$$\checkmark a(\phi) = 1, c(y, \phi) = 0$$



泊松回归 (Poisson regression)

$$f(y|\theta, \phi) = \exp\left(\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right)$$

□ 广义线性模型 (generalized linear model)

➤ 泊松分布 (Poisson distribution)

$$f(y) = \frac{e^{-\mu} \mu^y}{y!}$$

$$f(y) = \exp(y \log \mu - \mu - \log y!)$$

- ✓ $\theta = \log \mu$, 或 $\mu = e^\theta$
- ✓ $b(\theta) = \mu = e^\theta$
- ✓ $a(\phi) = 1$, $c(y, \phi) = -\log y!$



泊松回归 (Poisson regression)

□ 广义线性模型 (generalized linear model)

➤ 连接函数 g 如何选择?

- ✓ 正态分布: $g(\mu) = \eta = \mathbf{X}^T \boldsymbol{\beta} = \mu$
- ✓ 泊松分布? 指数族分布? 什么是 “正确的” g ?



泊松回归 (Poisson regression)

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➤ 标准连接函数 (canonical link)

- ✓ 设定 $\theta = \eta$



泊松回归 (Poisson regression)

□ 广义线性模型 (generalized linear model)

➤ 连接函数 g 如何选择?

- ✓ 正态分布: $g(\mu) = \eta = \mathbf{X}^T \boldsymbol{\beta} = \mu$
- ✓ 泊松分布? 指数族分布? 什么是 “正确的” g ?

➤ 标准连接函数 (canonical link)

- ✓ 设定 $\theta = \eta$
- ✓ 因此 $g(\mu) = \eta = \theta$

$$\begin{aligned} \text{伯努利分布: } g(\mu) &= \log\left(\frac{\mu}{1-\mu}\right) \\ \text{泊松分布: } g(\mu) &= \log \mu \end{aligned}$$

- ✓ 标准连接函数并非唯一 “正确的” g !



泊松回归 (Poisson regression)

□ 广义线性模型 (generalized linear model)

如何估计 GLM?



泊松回归 (Poisson regression)

□ 广义线性模型 (generalized linear model)

如何估计 GLM: 最大似然估计!

➤ 假设独立样本 $(x_i, y_i), i = 1, 2, \dots, n$, 每个 $y_i|x_i$ 均服从指数族分布

$$f(y_i|\theta_i, \phi) = \exp\left(\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\right), \quad i = 1, 2, \dots, n$$

✓ θ_i 随样本变化, ϕ 不变

• 线性回归 $y_i|x_i \sim N(\mu_i, \sigma^2)$

✓ 因为 $g(\mu_i) = \eta_i = x_i^T \boldsymbol{\beta}$, 所以 $\hat{\mu}_i = g^{-1}(x_i^T \hat{\boldsymbol{\beta}})$



泊松回归 (Poisson)

$$f(y|\theta, \phi) = \exp\left(\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right)$$

□ 广义线性模型 (generalized linear model)

➤ 构建似然函数

$$L(\theta) = \prod_{i=1}^n f(y_i|\theta_i, \phi)$$



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➤ 转化为对数似然函数

$$LL(\theta) = \sum_{i=1}^n \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + \sum_{i=1}^n c(y_i, \phi)$$



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➤ 最大化对数似然函数

✓ 和 β 有关吗?



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➤ 最大化对数似然函数

✓ **和 β 有关**: 标准连接函数 $\theta = \eta = X^T \beta$



泊松回归 (Poisson regression)

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常数!



泊松回归 (Poisson regression)

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➤ 等价于最大化

常数!

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泊松回归 (Poisson regression)

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- 最大化对数似然函数

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- 等价于最大化

常数!

$$LL(\theta) = \sum_{i=1}^n y_i \theta_i - b(\theta_i)$$

- 在标准连接函数 $\theta_i = \eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$ 的条件下

$$LL(\boldsymbol{\beta}) = \sum_{i=1}^n y_i \mathbf{x}_i^T \boldsymbol{\beta} - b(\mathbf{x}_i^T \boldsymbol{\beta})$$



泊松回归 (Poi

$$LL(\boldsymbol{\beta}) = \sum_{i=1}^n y_i \mathbf{x}_i^T \boldsymbol{\beta} - b(\mathbf{x}_i^T \boldsymbol{\beta})$$

□ 广义线性模型 (generalized linear model)

① 正态分布 $b(\theta) = \frac{\mu^2}{2} = \frac{\theta^2}{2}$

$$LL(\boldsymbol{\beta}) = \sum_{i=1}^n y_i \mathbf{x}_i^T \boldsymbol{\beta} - \frac{(\mathbf{x}_i^T \boldsymbol{\beta})^2}{2}$$

➤ 与线性回归最小化误差平方和等价吗?

泊松回归 (Poisson Regression)

$$LL(\boldsymbol{\beta}) = \sum_{i=1}^n y_i \mathbf{x}_i^T \boldsymbol{\beta} - b(\mathbf{x}_i^T \boldsymbol{\beta})$$

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最大化

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➤ 与线性回归最小化误差平方和等价!

最小化

$$\sum_{i=1}^n (y_i - \mathbf{x}_i^T \boldsymbol{\beta})^2$$

泊松回归 (Poi

$$LL(\boldsymbol{\beta}) = \sum_{i=1}^n y_i \mathbf{x}_i^T \boldsymbol{\beta} - b(\mathbf{x}_i^T \boldsymbol{\beta})$$

□ 广义线性模型 (generalized linear model)

② 泊松分布 $b(\theta) = \mu = e^\theta$

$$LL(\boldsymbol{\beta}) = \sum_{i=1}^n y_i \mathbf{x}_i^T \boldsymbol{\beta} - \exp(\mathbf{x}_i^T \boldsymbol{\beta})$$

➤ 泊松回归参数估计的对数似然函数



泊松回归 (Poisson regression)

□ 如何定量分析自变量的变化对因变量的影响？

➤ 弹性 (elasticity)

✓ 自变量变化1%，因变量变化百分率（无量纲）： $\frac{dy}{y} / \frac{dx}{x}$

✓ 弹性值-1.32：x 增加 1%，y 减少1.32%



泊松回归 (Poisson regression)

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➤ 边际效应 (marginal effect)

▪ 自变量变化1个单位，因变量变化量（有量纲）： $\frac{dy}{dx}$

▪ 吃一斤饭，长半斤肉 vs. 吃一公斤饭，长一斤肉（计量单位的“陷阱”）



泊松回归 (Poisson regression)

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➤ 弹性分析

① 连续变量

连接函数

$$g(\lambda_i) = \log \lambda_i = \eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$$



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$$\lambda_i = e^{\eta_i} = e^{\mathbf{x}_i^T \boldsymbol{\beta}}$$



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弹性定义

$$E_{x_{ik}}^{\lambda_i} = \frac{\frac{\Delta \lambda_i}{\lambda_i}}{\frac{\Delta x_{ik}}{x_{ik}}} = \frac{\Delta \lambda_i}{\Delta x_{ik}} \frac{x_{ik}}{\lambda_i} \approx \frac{\partial \lambda_i}{\partial x_{ik}} \frac{x_{ik}}{\lambda_i}$$



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链式法则

$$E_{x_{ik}}^{\lambda_i} = \frac{\partial \lambda_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial x_{ik}} \frac{x_{ik}}{\lambda_i} = \lambda_i \beta_k \frac{x_{ik}}{\lambda_i} = \beta_k x_{ik}$$



泊松回归 (Poisson regression)

□ 如何定量分析自变量的变化对因变量的影响？

➤ 弹性分析

② 名义尺度变量

把名义尺度变量转化为哑元变量，计算**伪弹性** (pseudo-elasticity)

连接函数

$$g(\lambda_i) = \log \lambda_i = \eta_i = x_i^T \beta$$



$$\lambda_i = e^{\eta_i} = e^{x_i^T \beta}$$

泊松回归 (Poisson regression)

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$$\lambda_i = e^{\eta_i} = e^{\mathbf{x}_i^T \boldsymbol{\beta}}$$

弹性定义



$$\begin{aligned} E_{x_{ik}}^{\lambda_i} &= \frac{\frac{\Delta \lambda_i}{\lambda_i}}{\frac{\Delta x_{ik}}{x_{ik}}} = - \frac{\Delta \lambda_i}{\lambda_i} = - \frac{e^{(x_{i1}\beta_1 + \dots + x_{i(k-1)}\beta_{k-1} + x_{i(k+1)}\beta_{k+1} + \dots)} - e^{(x_{i1}\beta_1 + \dots + \beta_k + \dots)}}{e^{(x_{i1}\beta_1 + \dots + \beta_k + \dots)}} \\ &= \frac{e^{\beta_k} - 1}{e^{\beta_k}} \end{aligned}$$

泊松回归 (Poisson regression)

□ 如何判别参数的显著性?

- 在一定条件下, 最大似然估计值 $\hat{\beta}$ 渐进于正态分布 (Wald test)

泊松回归 (Poisson regression)

□ 如何判别参数的显著性?

➤ 在一定条件下, 最大似然估计值 $\hat{\beta}$ 渐进于正态分布 (Wald test)

✓ 用 Z 或 t 统计量进行假设检验

✓ 什么时候 Z , 什么时候 t ?

泊松回归 (Poisson regression)

□ 如何判别参数的显著性?

➤ 在一定条件下, 最大似然估计值 $\hat{\beta}$ 渐进于正态分布 (Wald test)

✓ 用 Z 或 t 统计量进行假设检验

✓ 用 Z 或 t 取决于指数族概率密度或质量函数的 ϕ

- 线性回归 $\phi = \sigma$, 需估计 $\rightarrow t$
- 泊松回归 $a(\phi) = 1$, 无需估计 $\rightarrow Z$
- 数据量足够大, Z 和 t 的差别可忽略

泊松回归 (Poisson regression)

□ 如何评价模型拟合度 (goodness of fit)

① 似然比检验 (likelihood ratio test)

- 定量比较两个模型拟合度
- 似然比:

$$\lambda(\mathbf{X}) = \frac{\sup\{L(\boldsymbol{\theta}; \mathbf{X}): \boldsymbol{\theta} \in \Theta_0\}}{\sup\{L(\boldsymbol{\theta}; \mathbf{X}): \boldsymbol{\theta} \in \Theta\}}$$

- ✓ 给定观测数据集, 确定该数据集在 Θ_0 条件下发生的可能性, 以及在 Θ 条件下发生的可能性; 若两者比值 (必小于1) 较小, 则应拒绝 Θ_0



泊松回归 (Poisson regression)

□ 如何评价模型拟合度 (goodness of fit)

① 似然比检验 (likelihood ratio test)

- 定量比较两个模型拟合度
- 似然比:

$$\lambda(\mathbf{X}) = \frac{\sup\{L(\boldsymbol{\theta}; \mathbf{X}): \boldsymbol{\theta} \in \Theta_0\}}{\sup\{L(\boldsymbol{\theta}; \mathbf{X}): \boldsymbol{\theta} \in \Theta\}}$$

- ✓ 给定观测数据集, 确定该数据集在 Θ_0 条件下发生的可能性, 以及在 Θ 条件下发生的可能性; 若两者比值 (必小于1) 较小, 则应拒绝 Θ_0

- 似然比检验量:

$$\Lambda(\mathbf{X}) = -2 \log \lambda(\mathbf{X}) = 2 \left(L(\hat{\boldsymbol{\theta}}; \mathbf{X}) - L(\hat{\boldsymbol{\theta}}_0; \mathbf{X}) \right)$$



泊松回归 (Poisson regression)

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- 似然比检验量:

$$\Lambda(\mathbf{X}) = -2 \log \lambda(\mathbf{X}) = 2 \left(LL(\hat{\boldsymbol{\theta}}; \mathbf{X}) - LL(\hat{\boldsymbol{\theta}}_0; \mathbf{X}) \right)$$

样本足够

$$\Lambda(\mathbf{X}) \sim \chi^2(p), p = \dim(\boldsymbol{\theta})$$

$$\begin{aligned} H_0: \boldsymbol{\theta} &= \boldsymbol{\theta}_0 \\ H_1: \boldsymbol{\theta} &\neq \boldsymbol{\theta}_0 \end{aligned}$$



泊松回归 (Poisson regression)

□ 如何评价模型拟合度 (goodness of fit)

② McFadden' s pseudo R^2

$$R^2_{\text{McF}} = 1 - \frac{LL(\hat{\beta})}{LL(\mathbf{0})}$$

- $LL(\hat{\beta})$: 最大似然估计对应的对数似然函数值
- $LL(\mathbf{0})$: 参数均为0 (除了截距) 的对数似然函数值



泊松回归 (Poisson regression)

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- $LL(\mathbf{0})$: 参数均为0 (除了截距) 的对数似然函数值
- $R^2_{\text{McF}} \in [0,1]$
 - ✓ $R^2_{\text{McF}} = 1$: 完美拟合, 似然函数值为1, 对数似然函数值为0
 - ✓ $R^2_{\text{McF}} = 0$: 参数均为0 (除了截距), 自变量无法解释因变量的任何变化性



泊松回归 (Poisson regression)

□ 泊松回归——案例

Poisson Regression of Injury Accident Data

Independent Variable	Estimated Parameter	<i>t</i> Statistic
Constant	-0.826	-3.57
Average annual daily traffic on major road	0.0000812	6.90
Average annual daily traffic on minor road	0.000550	7.38
Median width in feet	- 0.0600	- 2.73
Number of driveways within 250 ft of intersection	0.0748	4.54

Number of observations	84
Restricted log likelihood (constant term only)	-246.18
Log likelihood at convergence	-169.25
Chi-squared (and associated <i>p</i> -value)	153.85 (<0.0000001)
R_p -squared	0.4792
G^2	176.5



泊松回归 (Poisson regression)

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Number of observations

84

Restricted likelihood

Log likelihood

Chi-squared

R_p -squared

G^2

$$\lambda_i = \exp(\mathbf{x}_i^T \boldsymbol{\beta}) = \exp \begin{pmatrix} -0.826 \\ +0.0000812 AADT1_i \\ +0.00055 AADT2_i \\ -0.06 MEDIAN_i \\ +0.0748 DRIVE_i \end{pmatrix}$$



泊松回归 (Poisson regression)

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计数回归

□ 学习目标

- 计数数据
- 泊松回归
- **负二项回归**
- 零膨胀泊松回归、负二项回归



负二项回归 (negative binomial regression)

□ 负二项分布

- **二项分布**: 重复 Bernoulli 实验成功概率为 p , 条件为总共实验 n 次, 随机变量为 n 次实验中成功实验次数 k
- **负二项分布**: 重复 Bernoulli 实验成功概率为 p , 条件为累计出现 r 次失败, 随机变量为成功实验次数 k
 - ✓ **负**: 站在失败的角度看成功

$$Pr(k; n, p) = Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$Pr(k; r, p) = Pr(X = k) = \binom{k + r - 1}{k} p^k (1 - p)^r$$



负二项回归 (negative binomial regression)

□ 负二项分布

➤ 概率质量函数推导

$$Pr(k; r, p) = Pr(X = k) = \binom{k+r-1}{k} p^k (1-p)^r$$

定义

$$\begin{aligned} \binom{k+r-1}{k} &= \frac{(k+r-1)!}{k! (r-1)!} \\ &= \frac{(k+r-1)(k+r-2) \dots (r)}{k!} \\ &= (-1)^k \frac{(-r)(-r-1) \dots (-r-k+1)}{k!} \\ &= (-1)^k \binom{-r}{k} \end{aligned}$$

组合数公式

$$Pr(k; r, p) = Pr(X = k) = (-1)^k \binom{-r}{k} p^k (1-p)^r$$



负二项回归 (negative binomial regression)

□ 负二项分布

➤ 概率质量函数特征

$$Pr(k; r, p) = Pr(X = k) = \binom{k + r - 1}{k} p^k (1 - p)^r$$

✓ 期望 $E(X) = \frac{rp}{1-p}$

✓ 方差 $Var(X) = \frac{rp}{(1-p)^2}$



负二项回归 (negative binomial regression)

□ 负二项分布

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✓ 期望 $E(X) = \frac{rp}{1-p}$

✓ 方差 $Var(X) = \frac{rp}{(1-p)^2}$

✓ $Var(X) = E(X) + \frac{E(X)^2}{r} = E(X) + \alpha E(X)^2 > E(X)$

• 方差大于期望：**过离散 (overdispersion)**

• $\alpha = \frac{1}{r}$ 为离散参数：**当 $r \rightarrow \infty$ ，负二项分布 $\rightarrow$ 泊松分布**



负二项回归 (negative binomial regression)

□ 负二项分布

➤ 概率质量函数变形

$$Pr(k; r, p) = Pr(X = k) = \binom{k+r-1}{k} p^k (1-p)^r$$

✓ 引入伽玛函数 (gamma function) : $\Gamma(n) = (n-1)!$

p 和 r 的函数

$$Pr(X = k) = \frac{(k+r-1)!}{k! (r-1)!} p^k (1-p)^r = \frac{\Gamma(k+r)}{k! \Gamma(r)} p^k (1-p)^r$$



负二项回归 (negative binomial regression)

□ 负二项分布

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$$Pr(X = k) = \frac{(k+r-1)!}{k! (r-1)!} p^k (1-p)^r = \frac{\Gamma(k+r)}{k! \Gamma(r)} p^k (1-p)^r$$

✓ 因为 $\mathbb{E}(X) = \mu = \frac{rp}{1-p}$, 所以 $p = \frac{\mu}{\mu+r}$, 以及 $\alpha = \frac{1}{r}$

μ 和 α 的函数

$$Pr(X = k) = \frac{\Gamma(k + \alpha^{-1})}{k! \Gamma(\alpha^{-1})} \left(\frac{\mu}{\mu + \alpha^{-1}} \right)^k \left(\frac{\alpha^{-1}}{\mu + \alpha^{-1}} \right)^{\alpha^{-1}}$$



负二项回归 (negative binomial regression)

□ 负二项回归

➤ 从 GLM 的角度定义负二项回归

✓ 随机部分

$$f(y_i | x_i, \alpha, \boldsymbol{\beta}) = \frac{\Gamma(y_i + \alpha^{-1})}{y_i! \Gamma(\alpha^{-1})} \left(\frac{\mu_i}{\mu_i + \alpha^{-1}} \right)^{y_i} \left(\frac{\alpha^{-1}}{\mu_i + \alpha^{-1}} \right)^{\alpha^{-1}}$$



负二项回归 (negative binomial regression)

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✓ 系统部分

$$\eta = \mathbf{X}^T \boldsymbol{\beta} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 \dots + \beta_d X_d$$



负二项回归 (negative binomial regression)

□ 负二项回归

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$$f(y_i | x_i, \alpha, \boldsymbol{\beta}) = \frac{\Gamma(y_i + \alpha^{-1})}{y_i! \Gamma(\alpha^{-1})} \left(\frac{\mu_i}{\mu_i + \alpha^{-1}} \right)^{y_i} \left(\frac{\alpha^{-1}}{\mu_i + \alpha^{-1}} \right)^{\alpha^{-1}}$$

✓ 系统部分

$$\eta = \mathbf{X}^T \boldsymbol{\beta} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 \dots + \beta_d X_d$$

✓ 连接函数

和泊松回归一样

$$g(\mu) = \eta = \log \mu \rightarrow \mu = e^\eta = e^{\mathbf{X}^T \boldsymbol{\beta}}$$



负二项回归 (negative binomial regression)

□ 负二项回归

➤ 从 GLM 的角度求解负二项回归：最大似然估计！

$$f(y_i | x_i, \alpha, \beta) = \frac{\Gamma(y_i + \alpha^{-1})}{y_i! \Gamma(\alpha^{-1})} \left(\frac{\mu_i}{\mu_i + \alpha^{-1}} \right)^{y_i} \left(\frac{\alpha^{-1}}{\mu_i + \alpha^{-1}} \right)^{\alpha^{-1}}$$

$$\max_{\alpha, \beta} L(\alpha, \beta | \mathbf{X}) = \prod_{i=1}^n f(y_i | x_i, \alpha, \beta)$$



负二项回归 (negative binomial regression)

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$$\max_{\alpha, \boldsymbol{\beta}} L(\alpha, \boldsymbol{\beta} | \mathbf{X}) = \prod_{i=1}^n f(y_i | \mathbf{x}_i, \alpha, \boldsymbol{\beta})$$

➤ 课外阅读

- ✓ “Negative binomial regression_1.pdf” : 对数似然函数及其偏导
- ✓ “Negative binomial regression_2.pdf” : 负二项与泊松的数学关联



负二项回归 (negative binomial regression)

□ 负二项回归——案例

Negative Binomial Regression of Injury Accident Data

Independent Variable	Estimated Parameter	<i>t</i> Statistic
Constant	-0.931	-2.37
Average annual daily traffic on major road	0.0000900	3.47
Average annual daily traffic on minor road	0.000610	3.09
Median width in feet	-0.0670	-1.99
Number of driveways within 250 ft of intersection	0.0632	2.24
Overdispersion parameter, α	0.516	3.09
Number of observations	84	
Restricted log likelihood (constant term only)	-169.25	
Log likelihood at convergence	-153.28	
Chi-squared (and associated <i>p</i> -value)	31.95	
	(<0.0000001)	



计数回归

□ 学习目标

- 计数数据
- 泊松回归
- 负二项回归
- **零膨胀泊松回归、负二项回归**



零膨胀泊松回归、负二项回归

□ 什么是“零膨胀”？



零膨胀泊松回归、负二项回归

□ 什么是“零膨胀”？

- 观测数据为0的比重较大



零膨胀泊松回归、负二项回归

□ 什么是“零膨胀”？

- 观测数据为0的比重较大

□ 为什么“零膨胀”？



零膨胀泊松回归、负二项回归

□ 什么是“零膨胀”？

- 观测数据为0的比重较大

□ 为什么“零膨胀”？

- 两种截然不同的原因：未观测到 vs. 很难观测到
- 例如，某路段车道宽、限速低、车辆少，发生事故的可能性极低（很难观测到），但是不排除重大人为事故（未观测到）



零膨胀泊松回归、负二项回归

□ 什么是“零膨胀”？

- 观测数据为0的比重较大

□ 为什么“零膨胀”？

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□ 如何处理“零膨胀”？



零膨胀泊松回归、负二项回归

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□ 如何处理“零膨胀”？

- 分类！



零膨胀泊松回归、负二项回归

□ 零膨胀泊松回归

- 假设很难观测到的概率为 p_i
- 那么 $1 - p_i$ 为服从泊松分布的概率



零膨胀泊松回归、负二项回归

□ 零膨胀泊松回归

- 假设很难观测到的概率为 p_i
- 那么 $1 - p_i$ 为服从泊松分布的概率
- 那么**服从泊松分布但未观测到**的概率为 $(1 - p_i) \frac{e^{-\lambda_i} \lambda_i^0}{0!} = (1 - p_i) e^{-\lambda_i}$



零膨胀泊松回归

$$f(y) = \frac{e^{-\mu} \mu^y}{y!}$$

□ 零膨胀泊松回归

- 假设很难观测到的概率为 p_i
- 那么 $1 - p_i$ 为服从泊松分布的概率
- 那么**服从泊松分布但未观测到**的概率为 $(1 - p_i) \frac{e^{-\lambda_i} \lambda_i^0}{0!} = (1 - p_i) e^{-\lambda_i}$

$$Pr(y_i = 0) = p_i + (1 - p_i) e^{-\lambda_i}$$

$$Pr(y_i = k) = (1 - p_i) \frac{e^{-\lambda_i} \lambda_i^k}{k!}, k = 1, 2, \dots$$



零膨胀

$$f(y_i|x_i, \alpha, \beta) = \frac{\Gamma(y_i + \alpha^{-1})}{y_i! \Gamma(\alpha^{-1})} \left(\frac{\mu_i}{\mu_i + \alpha^{-1}} \right)^{y_i} \left(\frac{\alpha^{-1}}{\mu_i + \alpha^{-1}} \right)^{\alpha^{-1}}$$

□ 零膨胀负二项回归

➤ 与零膨胀泊松回归原理一样!

$$Pr(y_i = 0) = p_i + (1 - p_i) \left(\frac{\alpha^{-1}}{\mu_i + \alpha^{-1}} \right)^{\alpha^{-1}}$$

$$Pr(y_i = k) = (1 - p_i) \frac{\Gamma(k + \alpha^{-1})}{k! \Gamma(\alpha^{-1})} \left(\frac{\mu_i}{\mu_i + \alpha^{-1}} \right)^k \left(\frac{\alpha^{-1}}{\mu_i + \alpha^{-1}} \right)^{\alpha^{-1}}, k = 1, 2, \dots$$

零膨胀泊松回归、负二项回归

□ 零膨胀 vs. 非零膨胀

➤ 什么时候应该用零膨胀?



零膨胀泊松回归、负二项回归

□ 零膨胀 vs. 非零膨胀

➤ 什么时候应该用零膨胀：假设检验！

➤ Vuong 检验

✓ 对每个观测点，首先计算

$$m_i = \ln \frac{f_1(y_i | \mathbf{x}_i)}{f_2(y_i | \mathbf{x}_i)}$$

▪ $f_1(y_i | \mathbf{x}_i)$ 和 $f_2(y_i | \mathbf{x}_i)$ 分别为模型1和2的概率质量或密度函数



零膨胀泊松回归、负二项回归

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▪ $f_1(y_i | \mathbf{x}_i)$ 和 $f_2(y_i | \mathbf{x}_i)$ 分别为模型1和2的概率质量或密度函数

✓ 然后计算 Vuong 统计量（近似服从标准正态分布）

$$V = \frac{\sqrt{n\bar{m}}}{\sqrt{\frac{1}{n} \sum_{i=1}^n (m_i - \bar{m})^2}}, \quad \bar{m} = \frac{1}{n} \sum_{i=1}^n m_i$$



零膨胀泊松回归、负二项回归

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$$m_i = \ln \frac{f_1(y_i|x_i)}{f_2(y_i|x_i)}$$

▪ $f_1(y_i|x_i)$ 和 $f_2(y_i|x_i)$ 分别为模型1和2

✓ 然后计算 Vuong 统计量（近似服从

若 f_1 和 f_2 分别是零膨胀负二项分布和负二项分布的概率质量函数，在 0.05 的显著性水平下：

- ① $V \geq 1.96$ ：零膨胀负二项回归
- ② $V \leq -1.96$ ：负二项回归
- ③ $-1.96 < V < 1.96$ ：不显著

$$V = \frac{\sqrt{n\bar{m}}}{\sqrt{\frac{1}{n} \sum_{i=1}^n (m_i - \bar{m})^2}}, \quad \bar{m} = \frac{1}{n} \sum_{i=1}^n m_i$$



计数回归

□ 模型总结

Model type	Advantages	Disadvantages
Poisson	Most basic model; easy to estimate	Cannot handle over- and under-dispersion; negatively influenced by the low sample-mean and small sample size bias
Negative binomial/ Poisson-gamma	Easy to estimate can account for over-dispersion	Cannot handle under-dispersion; can be adversely influenced by the low sample-mean and small sample size bias
Poisson-lognormal	More flexible than the Poisson-gamma to handle over-dispersion	Cannot handle under-dispersion; can be adversely influenced by the low sample-mean and small sample size bias (less than the Poisson-gamma), cannot estimate a varying dispersion parameter
Zero-inflated Poisson and negative binomial	Handles datasets that have a large number of zero-crash observations	Can create theoretical inconsistencies; zero-inflated negative binomial can be adversely influenced by the low sample-mean and small sample size bias



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The statistical analysis of crash-frequency data: A review and assessment of methodological alternatives

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ABSTRACT

Gaining a better understanding of the factors that affect the likelihood of a vehicle crash has been an area of research focus for many decades. However, in the absence of detailed driving data that would help improve the identification of cause and effect relationships with individual vehicle crashes, most researchers have addressed this problem by framing it in terms of understanding the factors that affect the frequency of crashes – the number of crashes occurring in some geographical space (usually a roadway segment or intersection) over some specified time period. This paper provides a detailed review of the key issues associated with crash-frequency data as well as the strengths and weaknesses of the various methodological approaches that researchers have used to address these problems.

negatively influenced by the low sample-mean and small sample size bias; no multivariate extensions to date
model with one state having a long-term mean to determine or evaluate the type of temporal a priori; results sensitive to missing values complex to implement; may not be easily transferable to other datasets
cannot handle under-dispersion; can be adversely influenced by the low sample-mean and small sample size estimation process; may not be easily transferable to other datasets
estimation process; requires formulation of matrix
estimation process; may not be easily transferable to other datasets
more detailed data than traditional crash-models; time-varying explanatory variables are handled
easily transferable to other datasets; results can be difficult to interpret
estimation process; may not be transferable to other datasets; work as black-boxes; may not have identifiable parameters

参考文献

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